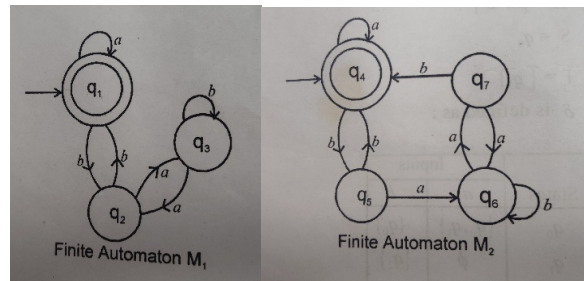


SSN College of Engineering
Department of Mathematics

Unit-1 Assignment

1. Construct a DFA that accepts all the strings on $\{0,1\}$ except those containing the substring 101.
2. Construct a non-deterministic finite automaton accepting the same set of strings over $\{a,b\}$ ending in aba. Use it to construct a DFA accepting the same set of strings.
3. Verify whether the finite automata M_1 and M_2 given below are equivalent over $\{a,b\}$



4. Construct a DFA that will accept strings on $\{a,b\}$ where the number of b's divisible by 3.
5. Construct a finite automaton that accepts the set of all strings in $\{a,b,c\}^*$ such that the last symbol in input string appears earlier in the string.
6. Consider the following ε -NFA. Compute the ε -closure of each state and find its equivalent DFA.

	ε	a	b	c
p	$\{q,r\}$	ϕ	$\{q\}$	$\{r\}$
q	ϕ	$\{p\}$	$\{r\}$	$\{p,q\}$
*r	ϕ	ϕ	ϕ	$\{r\}$

7. Design a Finite State Automata (FSA)

$L_1 = \{x \in \{0,1\}^* \mid x \text{ has even number of 0's and even number of 1's}\}$

$L_2 = \{x \in \{0,1\}^* \mid 0110 \text{ is a substring of } x\}$

$L_3 = \{x \in \{0,1\}^* \mid x \text{ represents a binary number divisible by } 3\}$

$L_4 = \{x \in \{0,1,2\}^* \mid \text{the sum of digits in } x \text{ is } 2 \text{ and } 3\}$

$L_5 = \{x \in \{0,1\}^* \mid \text{the number whose binary representation is } x \text{ and the number is } 2 \bmod 3\}$

$L_6 = \{x \in \{0,1\}^* \mid \text{the third bit from the left end is } 1 \text{ in } x\}$

$L_7 = \{x \in \{0,1\}^* \mid \text{the third bit from the right end is } 1 \text{ in } x\}$

8. Prove that if L is accepted by NFA with ϵ -transitions, then L is accepted by an NFA without ϵ -transitions.

9. Design a DFA and NFA for the language $L = \{ab \cup aba\}^*$.

10. Draw the state diagram for NFA accepting language

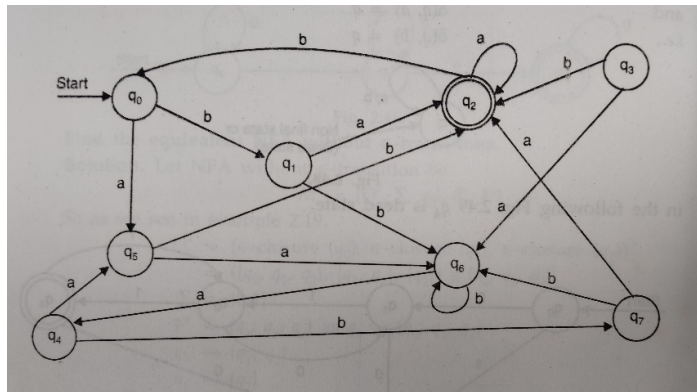
$$L = (ab)^* (ba)^* \cup aa^*$$

11. State and prove Pumping Lemma for Regular Languages.

12. Prove that Language $L = \{a^n b^n \text{ for } n = 0, 1, 2, 3, \dots\}$ is not regular.

13. Prove that Language $L = \{a^n b a^n \text{ for } n = 0, 1, 2, 3, \dots\}$ is not regular.

14. Minimize the DFA given as follows



15. Minimize the given DFA

